

Mithun Prabhu R

DevOps & Cloud Engineering Intern

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SUMMARY

DevOps & Cloud Engineering Intern candidate and CS Engineering student at Shiv Nadar University Chennai. Architected and maintained self-hosted private cloud environments, real-time network monitoring pipelines, and IoT telemetry systems. Adept at cross-functional problem-solving, communicating technical decisions clearly, and working independently to deliver production-grade infrastructure with Docker, Linux, Cloudflare Tunnel, and Python automation.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, SQL, HTML/CSS

DevOps / Infra: Docker, Docker Compose, Proxmox VE, LXC Containers, Nginx Reverse Proxy, Cloudflare Tunnel, Linux

Networking: Packet Sniffing (Scapy), Pi-hole DNS, MQTT, Network Traffic Analysis

Web & Backend: Flask, Node.js, React, MongoDB, MySQL, REST APIs

Tools & Workflow: Git, GitHub, Bash Scripting, Linux System Administration

Exploring: Kubernetes (K3s), Terraform, Prometheus, Grafana, GitHub Actions

PROJECTS

Home Lab & Private Cloud Infrastructure

Proxmox VE · Docker · Nginx · Cloudflare Tunnel · Linux

- Architected and provisioned a Proxmox VE hypervisor, independently orchestrating VMs and LXC containers to self-host production-grade services (Nextcloud, Jellyfin, Pi-hole) via Docker Compose.
- Designed and configured an Nginx reverse proxy for path-based service routing; led migration to Cloudflare Tunnel for zero-trust HTTPS exposure without opening inbound firewall ports.
- Managed 24/7 uptime through BIOS-level hardware tuning (CPU pinning, memory ballooning) and maintained version-controlled configs mirroring IaC practices, demonstrating ownership and attention to detail.

Wi-Fi Management System

Python · Flask · Scapy · MySQL · Chart.js

- Developed a 3-tier network monitoring platform using Scapy for real-time packet capture and per-device bandwidth analysis across a LAN.
- Engineered a Flask REST backend with MySQL to aggregate upload/download metrics; delivered a live Chart.js dashboard and automated daemon lifecycle via Bash on Ubuntu Server, communicating findings through clear visualisations.

Fleet-Sync — IoT Real-Time Tracking System

Python · MQTT · C · IoT Architecture

- Designed and implemented an MQTT publish-subscribe telemetry pipeline for low-latency GPS data transmission between C-based embedded firmware and a cloud monitoring station.
- Optimised geospatial data ingestion to handle high-frequency device updates, collaborating across hardware and software layers to ensure smooth real-time dashboard visualisation.

careerDrive

React · Node.js · MongoDB · Docker

- Developed and containerised a full-stack portfolio builder with authentication and dynamic milestone tracking; independently led the project from design to Docker deployment.

EDUCATION

B.Tech, Computer Science & Engineering — Shiv Nadar University

Chennai

CGPA: 8.9 / 10.0 (up to 3rd Semester)

Aug 2024 – Present

ACHIEVEMENTS

- Delivered end-to-end projects independently across embedded firmware, backend APIs, and self-hosted cloud infrastructure, demonstrating broad technical versatility.
- Designed and deployed a private cloud with Proxmox VE, Docker services, and Cloudflare Tunnel — demonstrating production-grade systems engineering beyond academics.
- Competed in hackathons and independently studied Kubernetes, Terraform, and observability tooling (Prometheus/Grafana) to deepen cloud-native DevOps proficiency.